Exponents & Exponential Functions Error Analysis

TEACHER KEY Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Product rule

ORIGINAL PROBLEM

Simplify $x^3 \cdot x^4$

INCORRECT WORKED SOLUTION

$x^3 \cdot x^4$

$= x^{3 \cdot 4}$

$= x^{12}$

A. EXPLAIN THE MISTAKE (1 POINT)

When multiplying powers with the same base, ADD the exponents: 3 + 4 = 7.

B. SHOW CORRECT WORK (1 POINT)

$x^3 \cdot x^4$

$= x^{3 + 4}$

$= x^7$

2. Power of a product

ORIGINAL PROBLEM

Simplify $(2x^3)^2$

INCORRECT WORKED SOLUTION

$(2x^3)^2$

$= 2x^6$

A. EXPLAIN THE MISTAKE (1 POINT)

The exponent 2 applies to EVERYTHING inside the parentheses, including the 2: $2^2 = 4$.

B. SHOW CORRECT WORK (1 POINT)

$(2x^3)^2$

$= 2^2 \cdot (x^3)^2$

$= 4x^6$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Quotient rule

ORIGINAL PROBLEM

Simplify $\frac{x^8}{x^2}$

INCORRECT WORKED SOLUTION

$\frac{x^8}{x^2}$

Divide the exponents: $\frac{8}{2} = 4$

$= x^4$

A. EXPLAIN THE MISTAKE (1 POINT)

When dividing powers with the same base, SUBTRACT the exponents: $8 - 2 = 6$.

B. SHOW CORRECT WORK (1 POINT)

$\frac{x^8}{x^2}$

$= x^{8-2}$

$= x^6$

4. Negative exponents

ORIGINAL PROBLEM

Evaluate 3^{-2}

INCORRECT WORKED SOLUTION

3^{-2}

$= -3^2$

$= -9$

A. EXPLAIN THE MISTAKE (1 POINT)

A negative exponent means a RECIPROCAL, not a negative number: $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$.

B. SHOW CORRECT WORK (1 POINT)

3^{-2}

$= \frac{1}{3^2}$

$= \frac{1}{9}$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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5. Zero exponent

ORIGINAL PROBLEM

Simplify $4x^0$ $(x \neq 0)$

INCORRECT WORKED SOLUTION

$4x^0$

$= 4 \cdot 0$

$= 0$

A. EXPLAIN THE MISTAKE (1 POINT)

Any nonzero base raised to the 0 power equals 1. So $x^0 = 1$, and $4x^0 = 4 \cdot 1 = 4$.

B. SHOW CORRECT WORK (1 POINT)

$4x^0$

$= 4 \cdot 1$

$= 4$

6. Negative base

ORIGINAL PROBLEM

Evaluate -3^2

INCORRECT WORKED SOLUTION

-3^2

$= (-3)(-3)$

$= 9$

A. EXPLAIN THE MISTAKE (1 POINT)

Without parentheses, the exponent applies only to 3: $-3^2 = -(3 \cdot 3) = -9$. (Only $(-3)^2$ equals 9.)

B. SHOW CORRECT WORK (1 POINT)

-3^2

$= -(3^2)$

$= -(3 \cdot 3)$

$= -9$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Exponential growth

ORIGINAL PROBLEM

A culture of 200 bacteria doubles every hour. Write a model and find the amount after 3 hours.

INCORRECT WORKED SOLUTION

$y = 200 \cdot 2t$

After 3 hours: $y = 200 \cdot 2(3)$

$y = 200 \cdot 6 = 1200$

A. EXPLAIN THE MISTAKE (1 POINT)

Doubling is repeated multiplication, so t belongs in the EXPONENT: $y = 200 \cdot 2^t$. After 3 hours: $200 \cdot 2^3 = 200 \cdot 8 = 1600$.

B. SHOW CORRECT WORK (1 POINT)

$y = 200 \cdot 2^t$

After 3 hours: $y = 200 \cdot 2^3$

$y = 200 \cdot 8 = 1600$

8. Decay factor

ORIGINAL PROBLEM

A $20,000 car loses 15% of its value each year. Find its value after 2 years.

INCORRECT WORKED SOLUTION

$y = 20{,}000(0.15)^t$

After 2 years: $20{,}000(0.15)^2$

$= 20{,}000(0.0225) = \$450$

A. EXPLAIN THE MISTAKE (1 POINT)

Losing 15% means KEEPING 85%, so the decay factor is $1 - 0.15 = 0.85$. Value = $20{,}000(0.85)^2 = 20{,}000(0.7225) = \$14{,}450$.

B. SHOW CORRECT WORK (1 POINT)

$y = 20{,}000(0.85)^t$

After 2 years: $20{,}000(0.85)^2$

$= 20{,}000(0.7225) = \$14{,}450$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. Multiplying terms

ORIGINAL PROBLEM

Simplify $(3x^2)(4x^3)$

INCORRECT WORKED SOLUTION

$(3x^2)(4x^3)$

$= 7x^5$

A. EXPLAIN THE MISTAKE (1 POINT)

Coefficients are MULTIPLIED ($3 \cdot 4 = 12$); only the exponents are added. The answer is $12x^5$.

B. SHOW CORRECT WORK (1 POINT)

$(3x^2)(4x^3)$

$= (3 \cdot 4)(x^2 \cdot x^3)$

$= 12x^5$

10. Linear or exponential?

ORIGINAL PROBLEM

Table: x = 0, 1, 2, 3 and y = 3, 6, 12, 24. Write an equation.

INCORRECT WORKED SOLUTION

Differences: 3, 6, 12

The pattern grows, so $y = 3x + 3$

Check x = 3: $y = 3(3) + 3 = 12$

A. EXPLAIN THE MISTAKE (1 POINT)

The y-values are MULTIPLIED by 2 each time (a constant ratio), so the table is exponential: $y = 3 \cdot 2^x$. The wrong equation does not even match the table (x = 3 should give 24).

B. SHOW CORRECT WORK (1 POINT)

Ratios: $\frac{6}{3} = \frac{12}{6} = \frac{24}{12} = 2$

Constant ratio, so $y = 3 \cdot 2^x$

Check x = 3: $3 \cdot 2^3 = 24$ ✓

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points